

Who and What....?



Rheumatoid Arthritis

Nov 17, 2024

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LECOM

Key Learning Objectives

- General rheumatic principles and some history
- Recognize clinical features, risk factors
- Recognize classification criteria for RA
- Pathophysiology and treatments

General Points

- Systemic autoimmune disease characterized by chronic inflammation (swelling, stiffness) of multiple joints
- ~1% of the population; F:M→3:1
- Genetic links similar to SLE

ORIGINAL ARTICLE

STAT4 and the Risk of Rheumatoid Arthritis and Systemic Lupus Erythematosus

Elaine F. Remmers, Ph.D., Robert M. Plenge, M.D., Ph.D., Annette T. Lee, Ph.D., Robert R. Graham, Ph.D., Geoffrey Hom, Ph.D., Timothy W. Behrens, M.D., Paul I.W. de Bakker, Ph.D., Julie M. Le, B.S., Hye-Soon Lee, M.D., Ph.D., Franak Batliwalla, Ph.D., Wentian Li, Ph.D., Seth L. Masters, Ph.D., et al.

September 6, 2007

N Engl J Med 2007; 357:977-986

- Loss of joint function, destruction of bone and cartilage
- Shorter life expectancy → systemic inflammation

**Paul Rubens: The Three
Graces (1638; Prado
Museum, Madrid, Spain).**



Appelboom T. Rheumatology 2005;44:681-683

RHEUMATOLOGY

Is RA a New World Disease?

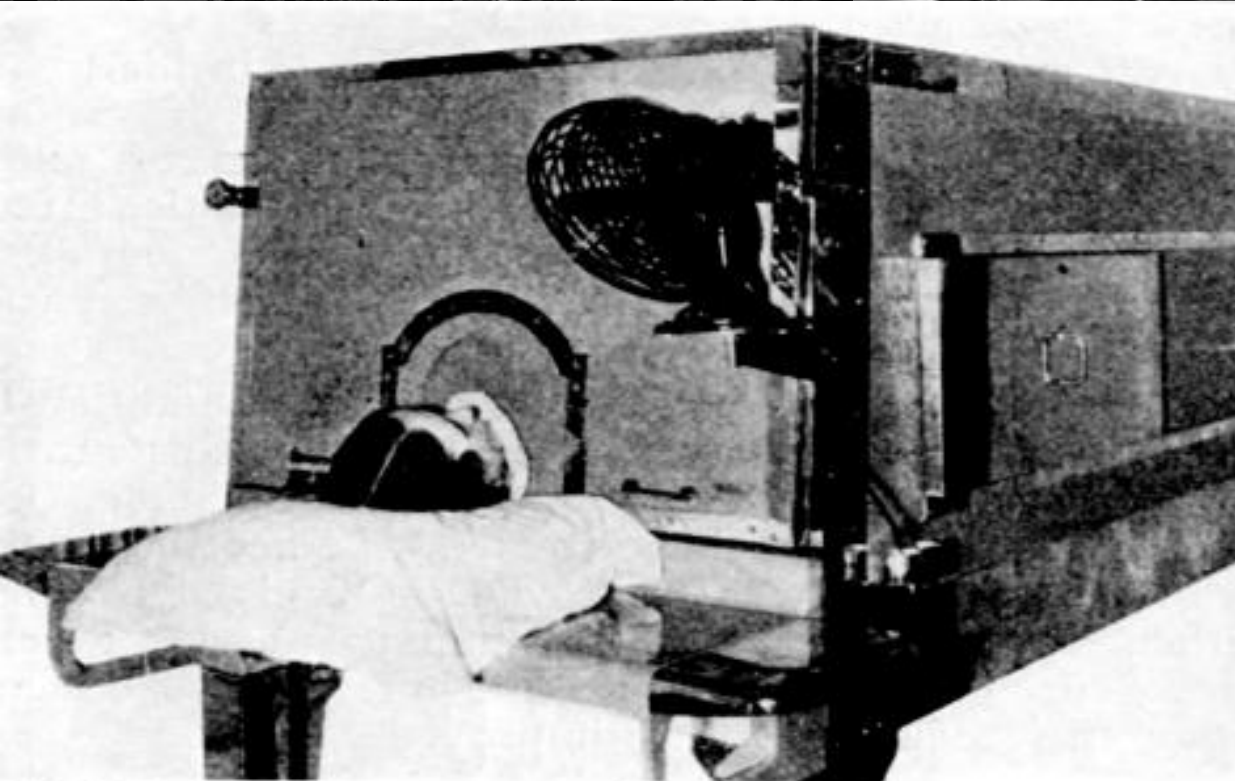
- Historical Epidemiology: Western view: “New World disease”
- General absence of radiographic changes in cohorts of plain films of prior populations
 - Gout, spondyloarthritis, osteoarthritis, DISH

Rothschild BM "Like a virgin": Absence of rheumatoid arthritis and treponematosi, good sanitation and only rare gout in Italy prior to the 15th century. Reumatismo. 2004

Appelboom T. Hypothesis: Rubens--one of the first victims of an epidemic of rheumatoid arthritis that started in the 16th-17th century? Rheumatology (Oxford). 2005

“New World” Epidemiology

- Environmental triggers (since Columbus?)
 - Industrialization
 - Crystalline silica dust
 - Automotive exhaust
 - Travel and connectivity (EBV, other agents?)
 - Refinement of lifestyle
 - Tobacco processing, use and distribution
 - Sugar refinement and ubiquity → gingival disease



Hunder GG, Matteson EL. Mayo Clin Proc. Apr 2010 A



Dr Philip Hench, 1950, Nobel Prize in medicine → synthetic cortisone

Case

CC: my fingers hurt

- Two 64 year old women present with similar symptoms: moderate aching and swelling in proximal interphalangeal joints for 2 yrs – worsening in past 3 months. One was initially episodic – lasting 3-5 days at time with periods of quiescence but now more chronic, the other gradually worsening over time.

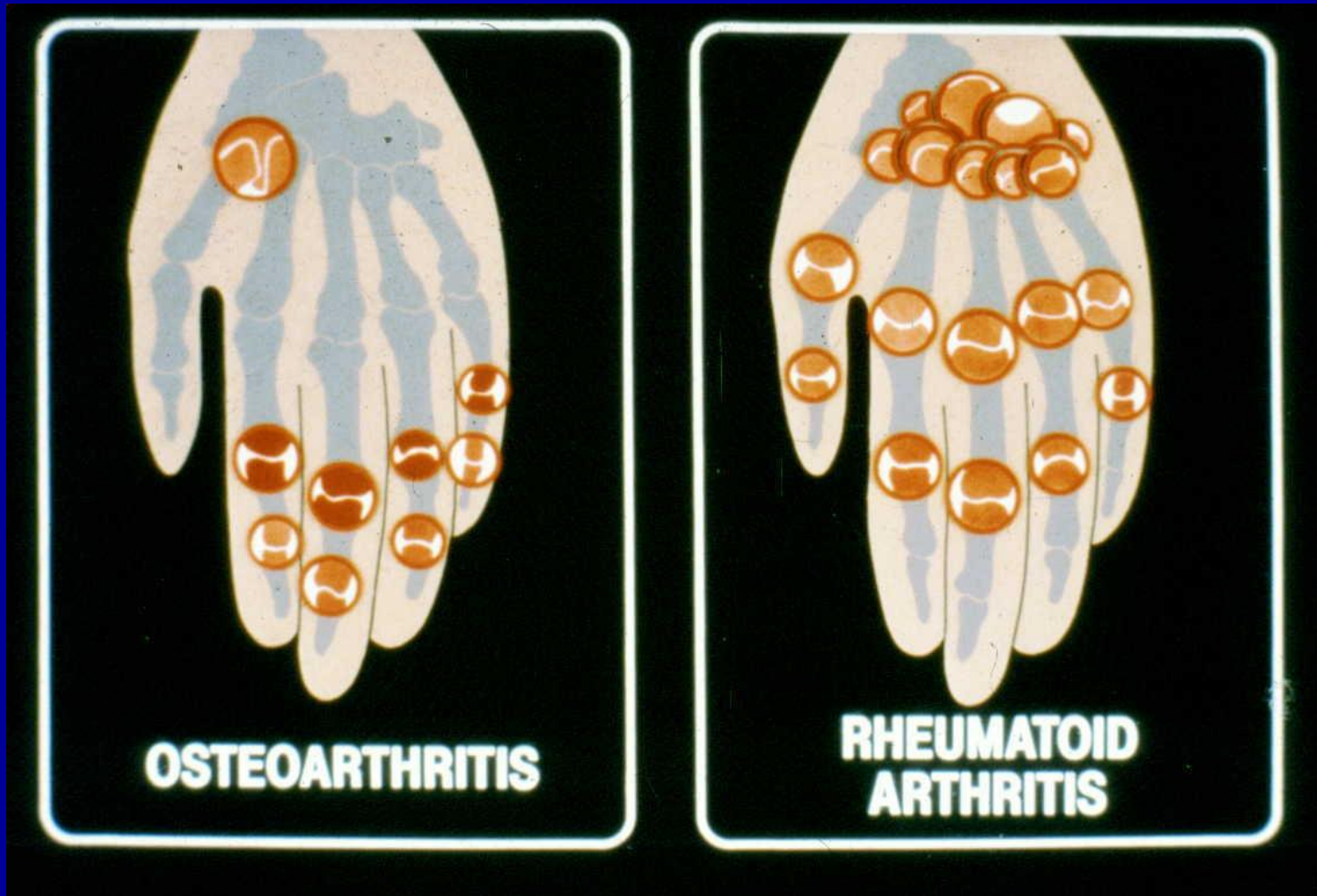
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- Which one is episodic?



Osteoarthritis vs. Rheumatoid Arthritis



Clinical Features



Courtesy ACR Image Bank

Growing evidence suggests RA pathogenesis involves bacteria in which part(s) of the body?

- Stomach/intestines
- Toenails/hair
- Larynx/lungs
- Skin
- Teeth/gingiva

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A

Phase A
Genetic
risk factors

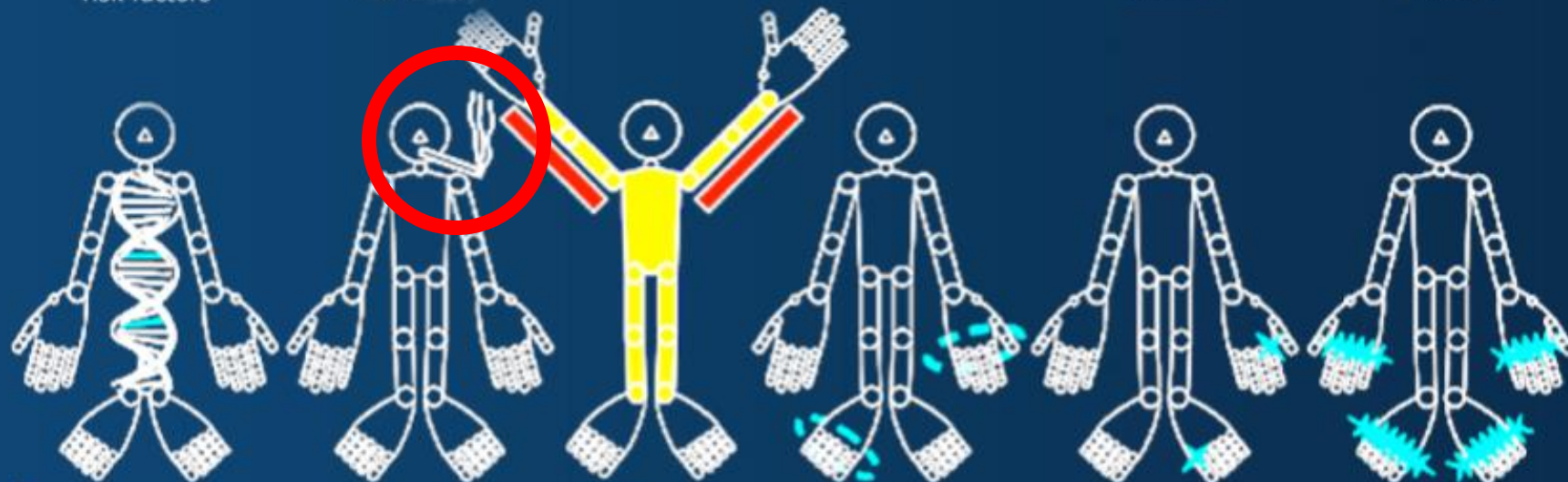
Phase B
Environmental
risk factors

Phase C
Autoimmunity

Phase D
Symptoms

Phase E
Undifferentiated
arthritis

Phase F
Rheumatoid
arthritis

**B**

Defining a set of healthy individuals

Identification of RA cases in predefined set

**C**

Identification of autoantibody positive arthralgia



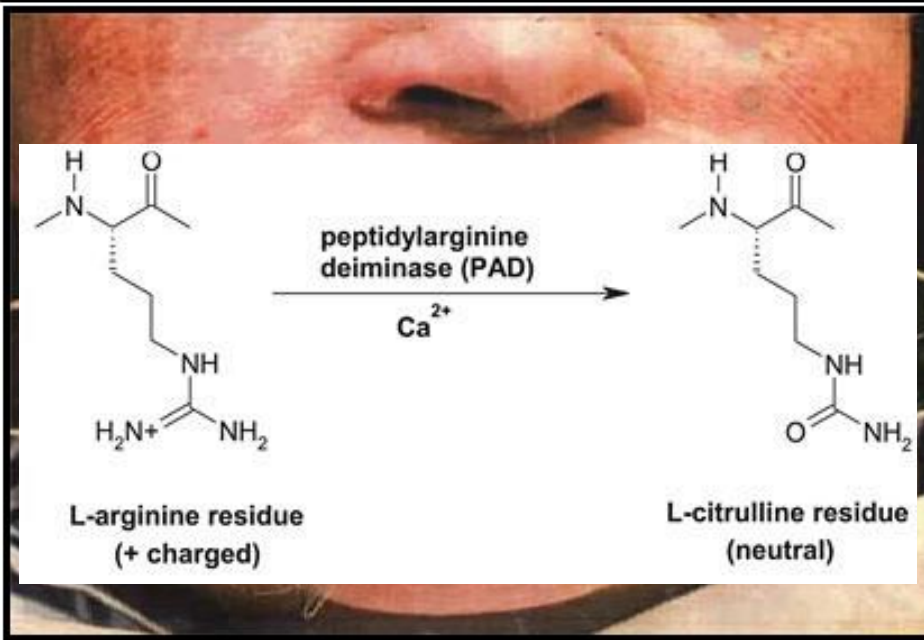
What auto antibody is most highly specific for the detection of RA?

- Anti-ribonucleoprotein (RNP)
- SSA
- Anti-cyclic citrillunated peptide (CCP)
- Rheumatoid factor (RF)
- Anti-tumor necrosis factor (TNF)

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Gingival Disease and Autoimmunity



- “Antibodies to citrullinated alpha-enolase peptide 1 are specific for Rheumatoid Arthritis and cross-react with porphyromonas gingivalis enolase.” Lundberg K., Kinloch A., Fisher, B.A., et al. *Arthritis Rheum.* 58: 3009-19 **2008**.
- “Associations of P. gingivalis titers with RF, antiCP suggests that infection with this organism plays a role in disease risk and progression in RA.” Mukuls TR, Payne, J.B., Reinhardt, R. A. et al *Int. Immunopharmacol.* 9:38-42 **2009**.
- “Antibodies to Porphyromonas gingivalis are Associated with Anticitrullinated Protein Antibodies in patients with Rheumatoid Arthritis and their Relatives.” Hitchon et al *J Rheumatol* 37:1105-1112 **2010**.

ACR 1987 Classification Criteria for Rheumatoid Arthritis

Patients Must Have Four of Seven Criteria:

Morning Stiffness Lasting at Least 1 Hour*

Swelling in 3 or More Joints*

Swelling in Hand Joints*

Symmetric Joint Swelling*

Erosions or Decalcification on X-ray of Hand

Rheumatoid Nodules

Abnormal Serum Rheumatoid Factor

* Must Be Present at Least 6 Weeks.

Real Case...

- 29 yo WF smoker with left wrist swelling and pain, right MTP swelling and pain x 9 yrs
- No stiffness or nodules or family history
- Previously diagnosed as overuse, tendonitis, 'degenerative wrist arthritis'
- Does she fulfill 1987 criteria?
- 3/7 classification criteria





Ref Rng

6/22/2011

CRP <10.0

20.0 (H)

ESR $0 - 20 \text{ MM/HR}$

23 (H)

CCP IGG <5.0

181.8 (H)

RF <11

<5

2010 ACR/EULAR RA classification criteria



RA can be classifiable or diagnosed with a score ≥ 6

JOINT DISTRIBUTION

1 large joint	0
2–10 large joints	1
1–3 small joints (large joints excluded)	2
4–10 small joints (large joints excluded)	3
>10 joints (at least 1 small joint)	5

SEROLOGY

Negative RF and negative ACPA	0
Low positive RF or ACPA ($\leq 3 \times$ ULN)	2
High positive RF or ACPA ($> 3 \times$ ULN)	3

SYMPTOM DURATION

<6 weeks	0
≥ 6 weeks	1

ACUTE PHASE REACTANTS

Normal CRP and ESR	0
Abnormal CRP or ESR	1

Case:

A 56 yo AA female presents with 6 mos of intermittent swelling and stiffness in hands that last 1-2 days and improves spontaneously and with ibuprofen. Bouts every 2 mos. None now

SH: teacher

½ ppd tob; no etoh or drugs

FH: mother with reported RA

Exam: no synovitis appreciable; no chronic changes

Work up?

Component	Latest Ref Rng	6/20/2016 
ESR	0 - 30 mm/Hr	20
CRP	<=7.4 mg/L	2.8 

“undifferentiated arthritis”

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Work up?

Component	Latest Ref Rng	6/20/2016 
ESR	0 - 30 mm/Hr	20
CRP	<=7.4 mg/L	2.8 
RF	<=14 IU/mL	190 (H)
CCP IGG	0.0 - 4.9 U/mL	>1200.0 (H)

Extra-articular manifestations

RA: Not Just a Joint Disease

CVD
10 Years Earlier

6-9x ↑
Serious Infections

2x ↑
Malignancy

Joint pain
Disability
Destruction

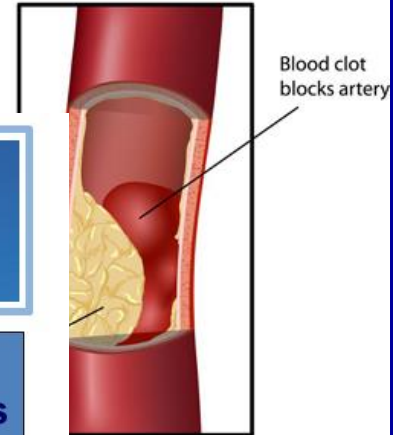
↑ Pulmonary
Disease

↑ GI
Bleeding

Life Expectancy Decreased
10 yr women, 4 yr men

Deane K. *J Musculoskel Med.* 2006;23:S24-S31.

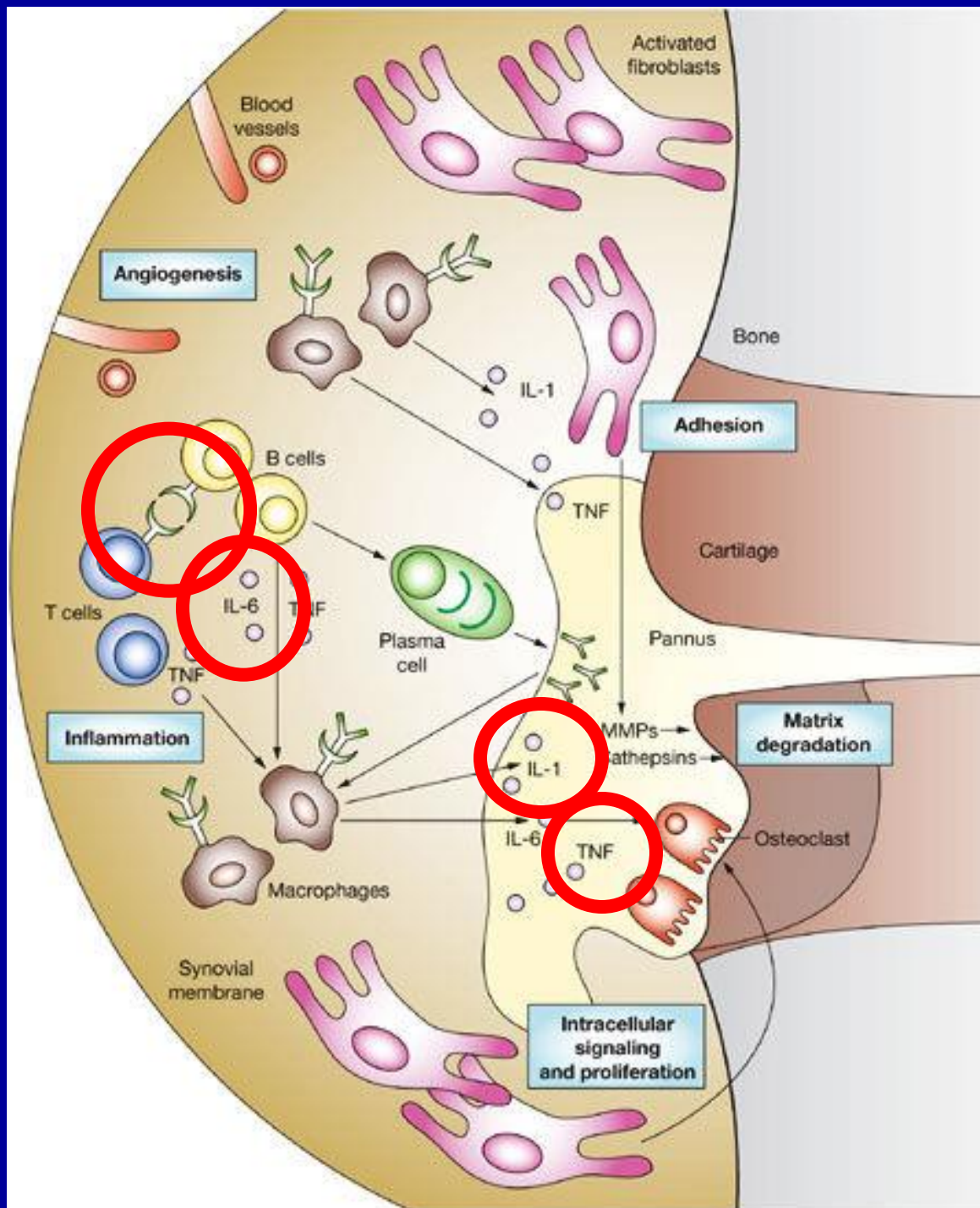
 **RheumNow**



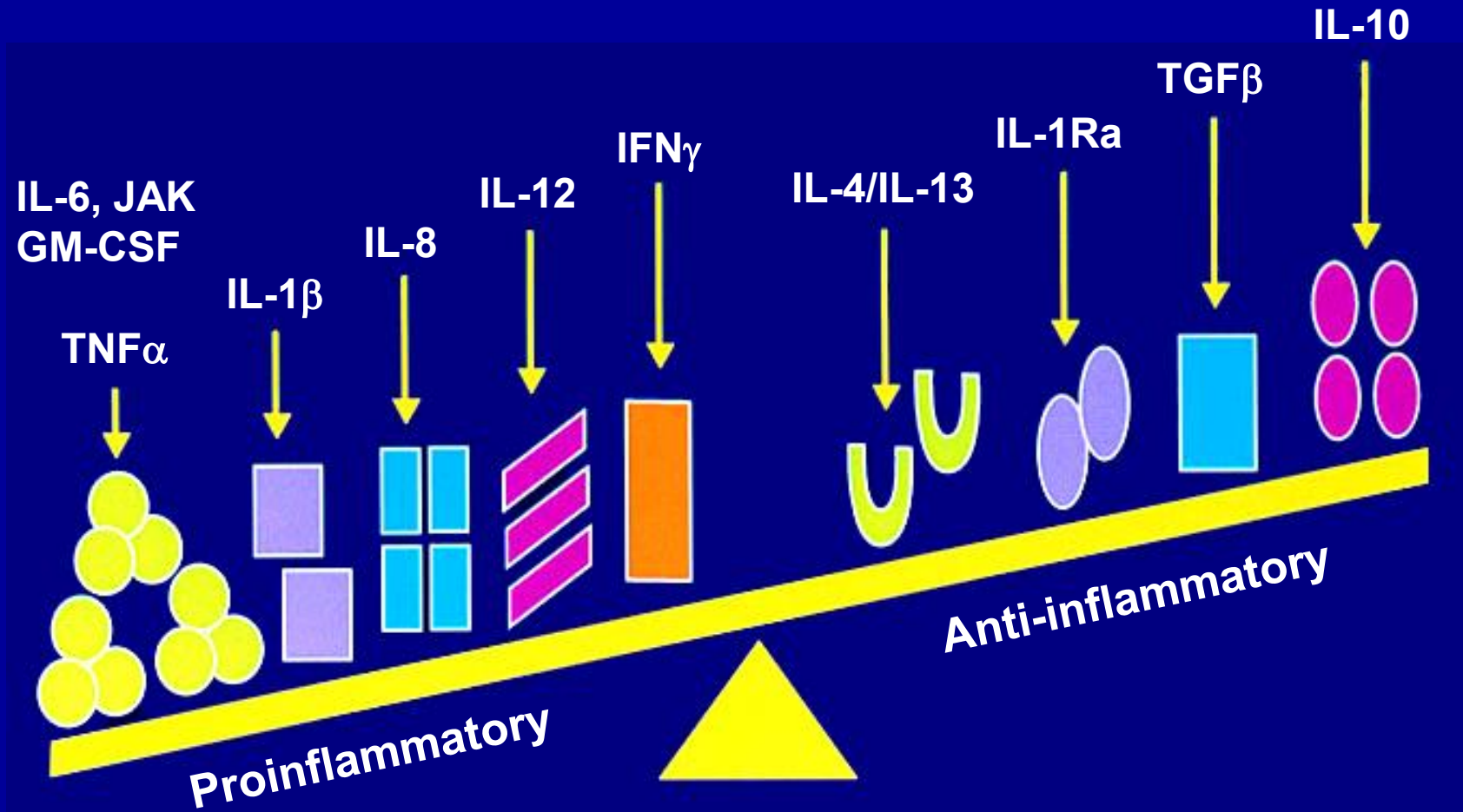
Treatment

Studies show that placebos work
just as well as antidepressants.
So...pick a hand.



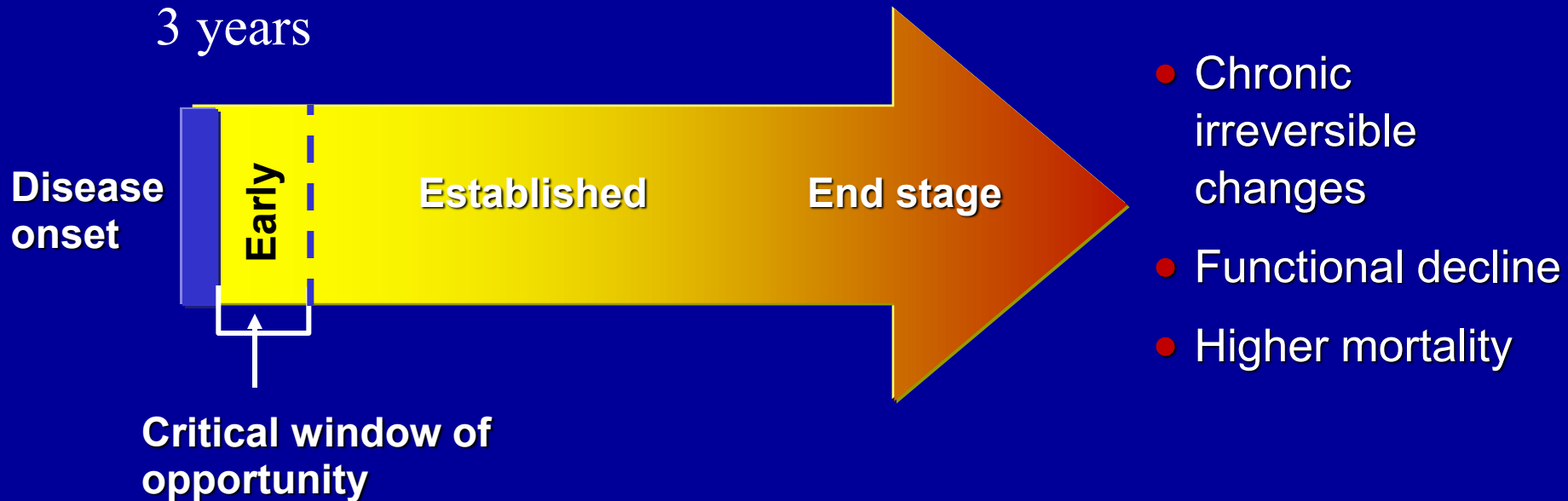


Chronic Inflammation: Imbalance Between Mediators



Window for Treating RA

- Radiographic progression occurs early and continues over the lifetime of a patient
- 70% of patients have radiographic damage within the first 3 years



RA Treatment: Optimizing Outcomes

- Early diagnosis and risk stratification
- DMARDs early (within 3 mo of symptoms)
- “TREAT TO TARGET”
- Steroids
 - bridge to effective DMARD therapy
 - prednisone >10 mg/d rarely needed for joints
 - minimize long term side effects (osteoporosis, DM)
- Co-morbidities
 - atherosclerosis
 - osteoporosis

Evolving RA Treatment Paradigm

Current Approach

Initial
treatment:
traditional
DMARDs

Add traditional
DMARD
(Combination
Therapy)

Evolving Paradigm

- Early aggressive treatment
- Biologics
 - Combination therapy

Which disease modifying anti-rheumatic drug (DMARD) is the most appropriate in an RA patient with chronic HCV?

- Leflunomide
- Methotrexate
- Sulfasalazine
- Azathioprine
- Hydroxychloroquine

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Which disease modifying anti-rheumatic drug (DMARD) is the most appropriate in an RA pt 10 wks gravid?

- Leflunomide
- Methotrexate
- Sulfasalazine
- Azathioprine
- Hydroxychloroquine

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Which disease modifying anti-rheumatic drug (DMARD) is the most appropriate for a 64 yo woman with breast cancer?

- tofacitinib
- adalimumab
- methotrexate
- rituximab

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- methotrexate
- tocilizumab
- anakinra

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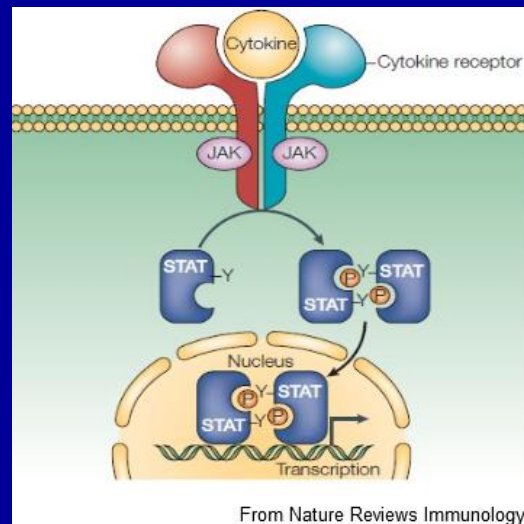
- leflunomide
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Traditional DMARD's

- methotrexate/Rheumatrex
- leflunomide/Arava
- sulfasalazine/Azulfidine
- azathioprine/Imuran
- hydroxychloroquine/Plaquenil
- mycophenolate mofetil/Cellcept
- gold
- minocycline
- doxycycline
- penicillamine
- cyclophosphamide
- cyclosporine

Biologic DMARD's

- TNF α antagonists:
 - etanercept (Enbrel)
 - infliximab (Remicade)
 - adalimumab (Humira)
 - golimumab (Simponi)
 - certolizumab (Cimzia)
 - **biosimilars**
- Interleukin-1 antagonist
 - anakinra (Kineret)
- Suppress T-Cell activation
 - abatacept (Orencia)
- Anti B-Cell monoclonal Ab
 - rituximab (Rituxan)
- Anti-interleukin-6
 - tocilizumab (Actemra)
 - sarilumab (Kevzara)
- Janus kinase (Jak) inhibitors
 - tofacitinib (Xeljanz)
 - baricitnib (Olumiant)
 - upadacitinib (Rinvoq)



Safety and Biologic DMARD's

- Serious Infections
- Opportunistic infections (TB, cocci, histo, blasto)
- Malignancy/lymphoma
- Demyelinating syndrome
- Hematologic abnormalities
- Congestive heart failure
- Hepatic
- Auto-antibodies and drug induced lupus
- Neutralizing antibodies

Safety and Biologic DMARD's

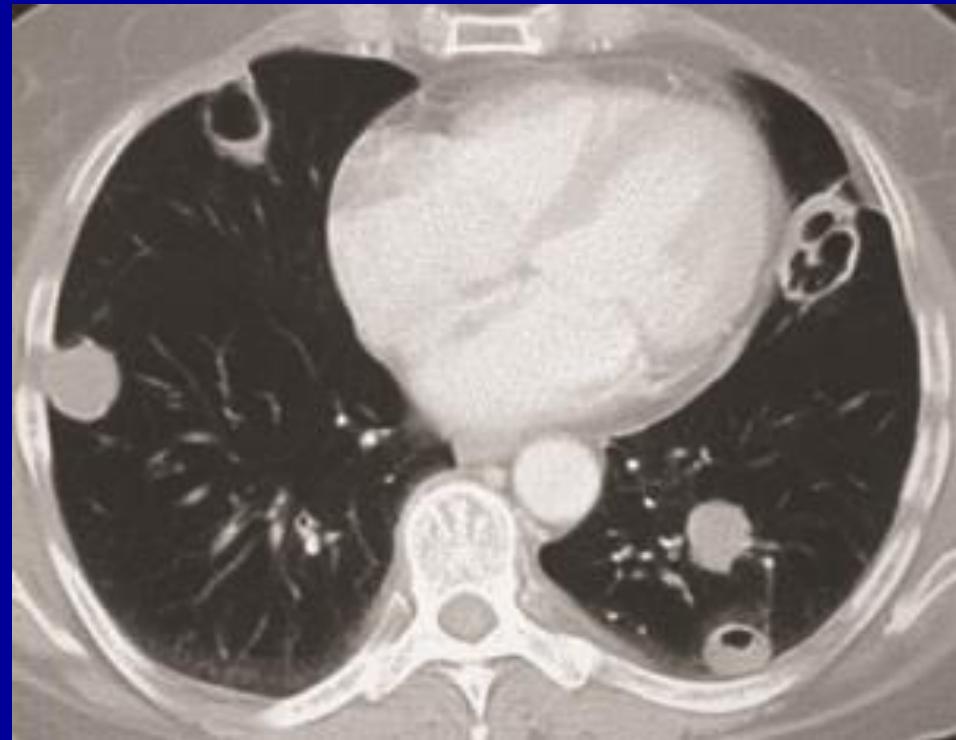
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Another case

“Glenn” is a 67 year old male musician with RA for 15 years but is hospitalized with dyspnea and is found to be hypoxic and needs to be intubated. Despite supportive care and aggressive measures for weeks, he succumbs. His manager suggests that his medications contributed to pneumonia.



RA: pulmonary manifestations



One Final Case.....





“The pain passes, but the beauty remains”

- Pierre-Auguste Renoir (1841-1919)

